

Project Brief: Predictive Modeling for Continuous Flow Reaction

1. System Architecture

The goal is to transition a **Batch Process** (time-dependent) to a **Continuous Flow Process**

2. Chemical Equation & Stoichiometry

The model operates on the following substitution reaction:



- **A (Organic Feed):** Primary substrate (MW: 161.03)
- **B (Aqueous Feed 1):** Nucleophile (MW: 45.09) at ~41% concentration.
- **C (Aqueous Feed 2):** Base (32% concentration).
- **P (Target Product):** Desired output (MW: 169.65).
- **S1, S2:** Byproducts (Water and Salt).

3. Training Datasets

- **Dataset A:** (R&D lab Time vs Concentration data)

Reaction: $A + B + C \rightarrow P + S1 + S2$

Molar Ratio: A : 1 mol eq, B : 1.3 mol eq wrt A, C : 1.2 mol eq wrt A

- Yield: 96.29%
- Selectivity of P: 96.40%
- Conversion of A: 99.89%
- Wt Fraction: 95 to 96 %

Temperature	Maintaining Time	A	P	Impurities
30-35°C	0hr	12.57%	84.04%	3.39%
	2hrs	2.22%	92.99%	4.79%
	4hrs	0.50%	94.49%	5.01%
	6hrs	0.15%	95.63%	4.22%
35-40°C	0hr	8.36%	87.95%	3.69%
	2hrs	1.83%	92.65%	5.52%
	4hrs	0.43%	94.80%	4.77%
40-42°C	0hr	8.23%	82.16%	9.61%
	2hrs	0.45%	90.40%	9.15%

	5hrs		91.14%	8.86%
45-50°C	0hr	6.31%	81.36%	12.33%
	2hrs	0.08%	92.29%	7.63%
	10hrs	0.06%	93.18%	6.76%
25-30°C	0hr	15.21%	81.50%	3.29%
	2hrs	4.19%	91.60%	4.21%
40-45°C	2hrs	0.11%	95.35%	4.54%

- **Dataset B: (Pilot Batches Data)**

Pilot Batches Data					
Component	Description	B#002	B#003	B#004	B#005
A	Qty (kg)	70	70	70	70
	Purity %	99.65%	99.65%	99.65%	99.65%
B	Qty (kg)	62	62.5	62.5	62
	Purity %	41.44%	41.19%	41.19%	41.19%
	Mole ratio wrt A	1.31	1.31	1.31	1.30
C	Qty (kg)	66	66	66	66
	Purity	31.50%	31.50%	31.50%	31.50%
	Mole ratio wrt A	1.20	1.20	1.20	1.20
A : RM Addition time (hrs)		2	2	2.5	2.5
Total Maintenance time (A/F Addition)		11	5	4	5
Mass Balance		98.89%	98.54%	98.82%	99.55%
A: Conversion %		99.99%	99.96%	99.92%	99.92%
P : Selectivity %		98.75%	98.60%	98.65%	98.65%
Reaction yield %		98.74%	98.56%	98.57%	98.57%

4.Validation Criteria

- The model shall be validated using laboratory-scale and pilot batch datasets to ensure robust prediction of output variables (Y) as a function of defined input variables (X).
- The model is expected to demonstrate high predictive accuracy and consistency across varying operating conditions (input variables), enabling identification of optimal process parameters to achieve target output variables for **continuous flow reaction**.

Key Validation Metrics

The model must meet the following performance thresholds:

- **Conversion Efficiency:**
Predict and sustain conversion of reactant A to product P at $\geq 99.8\%$
- **Reaction Yield:**
Accurately predict and achieve product yield $\geq 96.5\%$
- **Selectivity:**
Maintain selectivity towards desired product P at $\geq 96.8\%$, minimizing side reactions and impurity formation